

SALES & PROFIT ANALYSIS USING TABLEAU

1. INTRODUCTION

This project performs a comprehensive analysis of retail sales data using Tableau. The primary objective is to uncover strategic insights related to sales performance, profitability, customer behavior, product contribution, and shipping/order trends. By using Tableau's advanced visualization capabilities—filters, calculated fields, dashboards, and interactive charts—the project transforms raw data into actionable intelligence for business decision-making.

The dataset used for this study is publicly available on Kaggle:

<https://www.kaggle.com/datasets/anuvagoval/sales-store-product-details>

2. PROJECT OBJECTIVES

The project focuses on the following key analytical goals:

2.1 Sales & Profit Analysis by Customer Segment

- Compute total sales and profit for each segment.
- Identify the most profitable customer groups.
- Support segmentation-based marketing strategies.

2.2 Order Priority Analysis

- Evaluate how order priority levels (High, Medium, Low, Not Specified) impact sales and profit.
- Determine if priority levels reflect business importance or product urgency.

2.3 Profitability Relative to Sales

- Create a calculated field to measure profit-to-sales ratio.
- Identify high-margin products that drive the most profitability.

2.4 Average Order Value (AOV)

- Calculate revenue per order across product categories.
- Insights support pricing and bundling decisions.

2.5 Discount Impact on Profit

- Analyze how discounts influence profitability.
- Find optimal discount levels that maximize revenue without hurting margins.

3. METHODOLOGY

The analysis was performed using Tableau with the following steps:

3.1 Data Preparation

- Loaded dataset from Kaggle.
- Cleaned missing values, standardized fields.
- Created essential calculated fields:
 - Profit Ratio = Profit / Sales
 - Discount Impact = (Sales – Discounted Value) relative to Profit
 - Average Order Value = Sales / Quantity

3.2 Visualization Development

Four dashboards were created:

1. Sales & Profit Overview
2. Customer & Segment Analysis
3. Product Performance Analysis
4. Shipping & Order Priority Insights

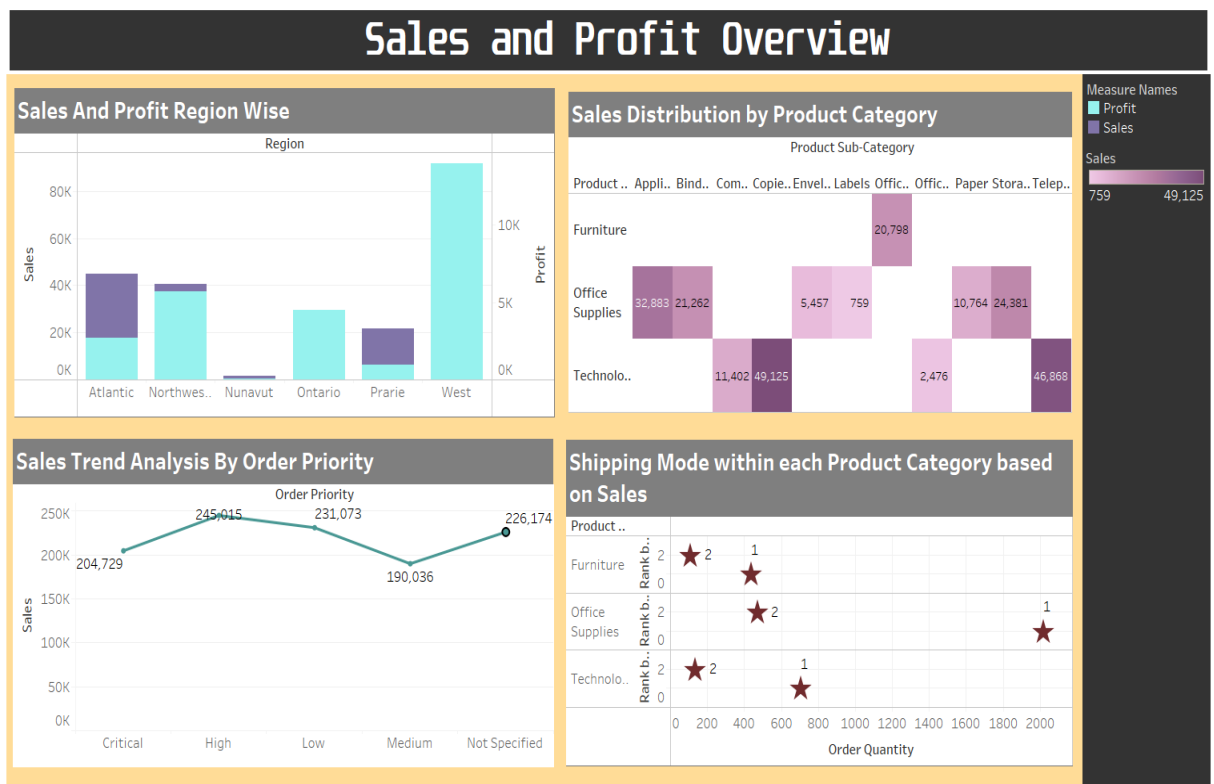
Each dashboard includes multiple interlinked charts for interactive exploration.

4. Dashboard Summaries & Insights

4.1 Dashboard 1: Sales and Profit Overview

Charts Used

1. **Bar Chart:** Sales vs Profit by Region
2. **Heat Map:** Sales Distribution by Product Category
3. **Line Chart:** Sales Trend Across Order Priority Levels
4. **Bump Chart:** Sales Ranking of Shipping Modes within Categories



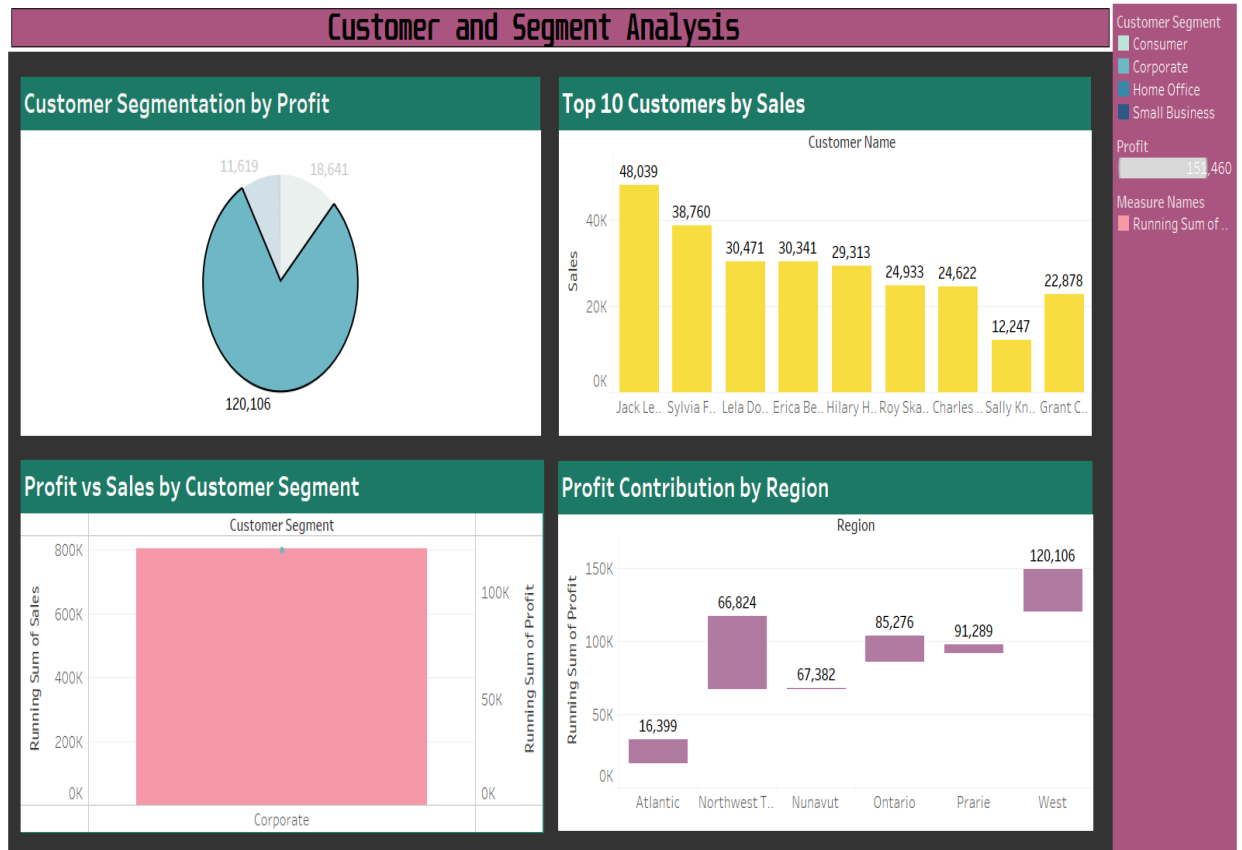
Insights

- **Region-wise Profitability:** Certain regions outperform others in both sales volume and profitability.
- **Product Category Heatmap:** Technology and Furniture show high sales density.
- **Order Priority Trends:** High-priority orders generally show steady sales with spikes during peak seasons.
- **Shipping Mode Bump Chart:** "Standard Class" typically dominates, but "Express Ship" rises within high-value categories.

4.2 Dashboard 2: Customer and Segment Analysis

Charts Used

1. **Pie Chart:** Profit Share by Customer Segment
2. **Horizontal Bar Chart:** Top 10 Customers by Sales
3. **Pareto Chart (80/20 Rule):** Customer Segment Contribution
4. **Waterfall Chart:** Profit Contribution by Region



Insights

- Corporate customers generate the highest overall profit.
- The top 10 customers contribute a significant portion of revenue—validating **Pareto principle**.
- Regional profits vary widely; some regions contribute strongly while others drag due to operational costs or discount levels.

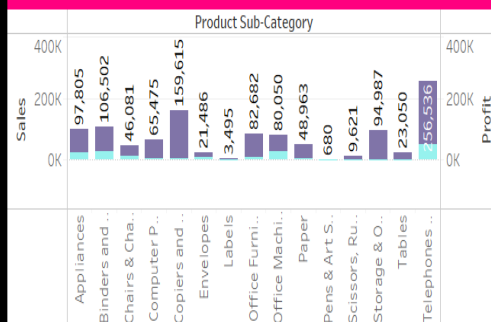
4.3 Dashboard 3: Product Performance Analysis

Charts Used

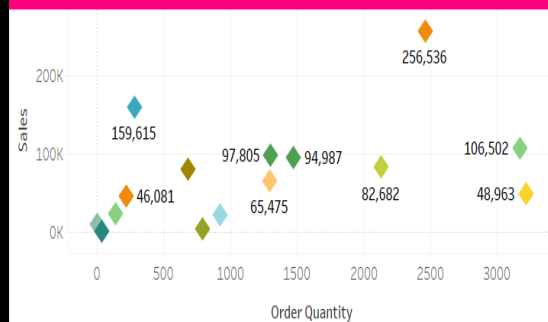
1. **Stacked Bar Chart:** Sales & Profit for Product Sub-Categories
2. **Waterfall Chart:** Category Contribution to Profit Margin
3. **Scatter Plot:** Sales vs Quantity (Sub-Categories)
4. **Histogram:** Sales Distribution

Product Performance Analysis

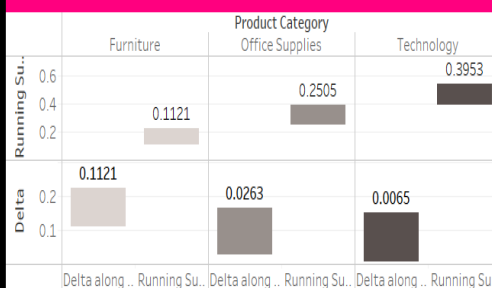
Sales and Profit by Product Sub-Category



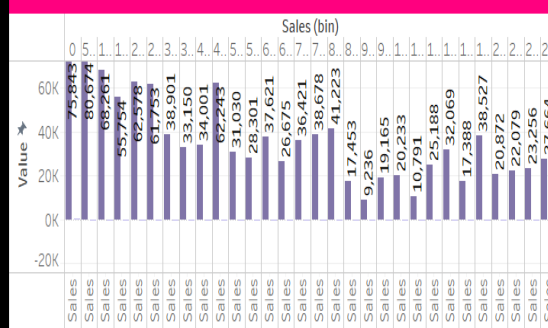
Sales vs Order Quantity by Product Sub-Category



Analyze the impact of each product category on overall profit margin



Distribution of Sales Amounts



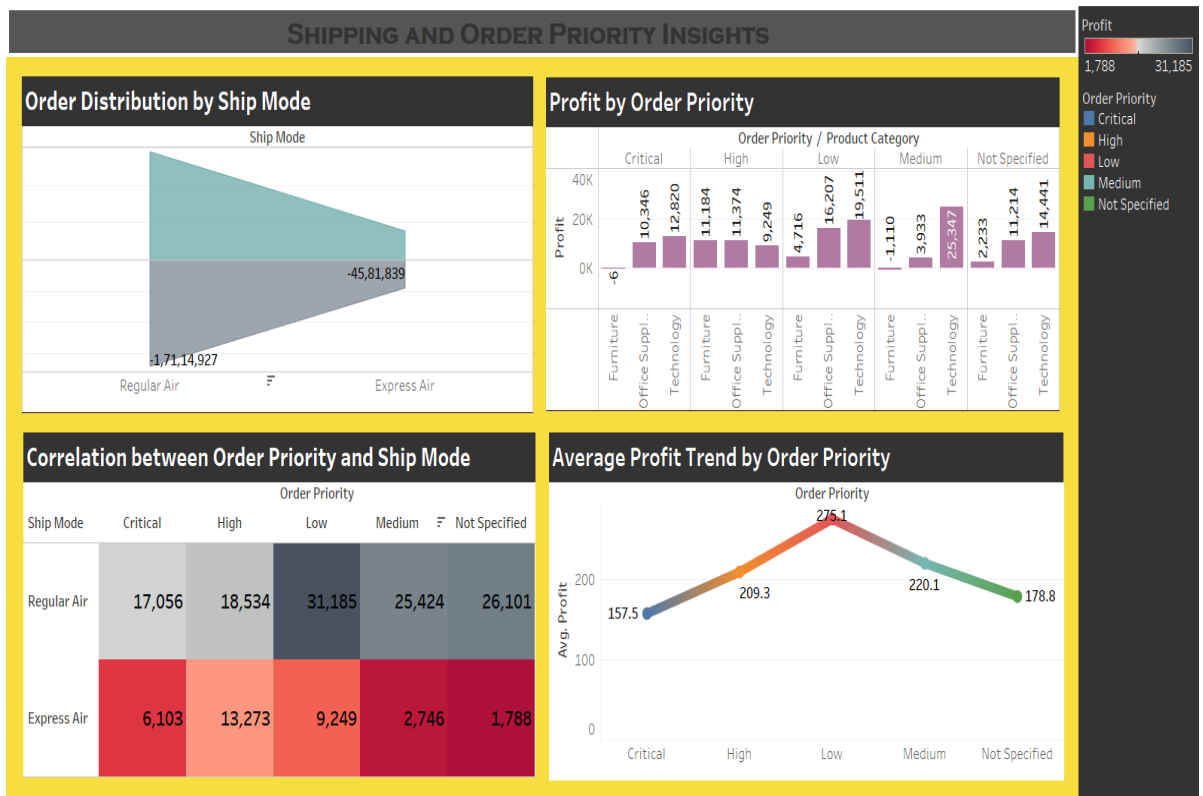
Insights

- Some sub-categories (e.g., Phones, Chairs) have high sales but variable profit margins.
- Waterfall chart identifies categories with negative profit contribution.
- Scatter plot shows that high quantity doesn't always mean high sales; price and discount influence total value.
- Sales distribution reveals skewness, with a high number of low-value orders.

4.4 Dashboard 4: Shipping and Order Priority Insights

Charts Used

- Funnel Chart:** Order Distribution by Ship Mode
- Grouped Bar Chart:** Profit by Order Priority
- Heat Map:** Order Priority vs Ship Mode
- Line Chart:** Profit Trend by Order Priority



Insights

- Standard shipping dominates overall order volume (funnel shaped).
- High-priority orders contribute more profit per transaction despite lower count.
- Heatmap shows strong alignment between “Express Ship” and higher priority orders.
- Profit trend line indicates that priority level directly influences profitability.

5. KEY FINDINGS

Sales Patterns

- Technology category leads in overall sales.
- Corporate and Consumer segments generate stable revenue.

Profitability

- Some high-sales products show low profit margins due to discounts.
- Profit-to-sales ratio reveals underperforming yet high-selling products.

Customer Insights

- A small portion of customers drive majority of sales (Pareto rule).

Order & Shipping Behavior

- Standard shipping is most used, but Express shipping yields higher profitability in many cases.
- Order Priority significantly affects profit trends.

6. BUSINESS RECOMMENDATIONS

1. Optimize Discount Strategy

- Reduce discount levels on high-selling but low-profit products.

2. Focus on High-Value Customer Segments

- Prioritize Corporate clients for targeted marketing.

3. Improve Logistics for High-Priority Orders

- Strengthen Express and Second Class shipping reliability.

4. Promote High-Margin Products

- Leverage insights from profit ratio analysis to boost profitable categories.

5. Expand in High-Performing Regions

- Based on regional profit mapping, invest more in high-performing markets.

7. CONCLUSION

This Tableau project transforms raw retail data into meaningful insights that guide strategic business decisions. The dashboards highlight patterns in profitability, customer behavior, sales performance, and shipping efficiency. With actionable insights on pricing, discounts, customer segmentation, and operational logistics, businesses can significantly improve their revenue and profitability.

This report provides a strong analytical foundation for retail sales optimization using visual analytics.